

Ama A. Koranteng

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<https://amakora0.github.io>

EDUCATION	Johns Hopkins University June 2025 PhD , Computer Science — <i>Advisor: Michael Dinitz</i>
	Johns Hopkins University Dec 2022 MSE , Computer Science
	Massachusetts Institute of Technology Feb 2020 BS , Mathematics
RESEARCH EXPERIENCE	Johns Hopkins University — Doctoral Researcher Sep 2020–present - Developed algorithms for graph fault-tolerance and minimization optimization problems, resulting in 4 publications: arXiv:2206.12245, arXiv:2304.06656, arXiv:2502.06522, arXiv:2502.10890 - Communicated research results to diverse audiences at 5 international and domestic conferences and seminars - Taught, mentored, and tutored high school, undergraduate, and junior PhD students in algorithms research, Python programming, and career development
	MIT Theory of Computation Group — Undergraduate Researcher Jun 2018–May 2019 <i>Advisor: Dr. Jayson Lynch</i> - Read and synthesized papers on gadget abstractions used to prove computational complexity of various games - Rigorously defined and reconciled new and existing frameworks and gadget abstractions - Solved open problems related to these papers, including designing algorithms and proving hardness results
WORK EXPERIENCE	Boeing — Applied Mathematics Intern Jun–Sep 2024 <i>Supervisor: Dr. Richard Joel Thompson</i> - Supported development of quantum heterogeneous computing resource estimator tool - Wrote software to take in Python-implemented algorithms and output directed acyclic graph (DAG) representations of the input algorithms - Developed mathematical mixed-integer linear program formulation for algorithm workflow optimization, based on the DAG representations
	Northrop Grumman — Intern Jun–Aug 2017 <i>Supervisor: Dr. Jonathan Las Fargeas</i> - Studied and modified existing distributed algorithms for collaborative drones - Implemented algorithms in C# using behavior trees in the Unity game engine
	Northrop Grumman — Intern Jun–Aug 2016 <i>Supervisor: Dr. Jonathan Las Fargeas</i> Created MATLAB models and Java programs to model complex offensive and defensive aircraft combat maneuvers
PUBLICATIONS	with Michael Dinitz and Yasamin Nazari Approximation Algorithms for Optimal Hopsets <i>International Colloquium on Automata, Languages, and Programming (ICALP), 2025</i>

with Greg Bodwin, Michael Dinitz, and Lily Wang

Light Edge Fault Tolerant Graph Spanners

International Colloquium on Automata, Languages, and Programming (ICALP), 2025

with Michael Dinitz, Guy Kortsarz, and Zeev Nutov

Improved Approximations for Relative Survivable Network Design

Workshop on Approximation and Online Algorithms (WAOA), 2023

with Michael Dinitz and Guy Kortsarz

Relative Survivable Network Design

International Conference on Approximation Algorithms for Combinatorial Optimization Problems (APPROX), 2022

AWARDS

NSF Graduate Research Fellowship (acceptance rate $\sim 15\%$)	2022
GEM Fellowship (declined) (acceptance rate $\sim 10\%$)	2022
Graduate Fellowship for STEM Diversity (declined)	2022
Wu and Zhang Endowed Graduate Student Fellowship	2020
Google Computer Science Research Mentorship Program	2019
Bloomberg Grace Hopper Travel Grant	2016
Generation Google Scholarship (acceptance rate $\leq 1\%$)	2014
NCWIT Aspirations in Computing Award	2014

SERVICE

Tutor , Intro to Algorithms	Oct 2024 - present
Teach 1-on-1 interactive lessons on introductory concepts in algorithms and discrete mathematics	
Mentor , Institute for African-American Mentoring in CS (IAAMCS)	Oct 2024 - present
Provide guidance to African-American computer science undergraduate students on applying to PhD programs, conducting research, and navigating obstacles	
Advisor and Mentor , JHU Whiting Internships in Science and Engineering	Jun 2023 - May 2024
- Advised and mentored a Baltimore city high school student - Introduced student to basic graph theory and algorithms concepts - Guided student through a programming project in which they implemented graph algorithms in Python - Mentored several other students in the WISE program	
Conference Subreviewer	
- ICALP 2025 - STACS 2025 - ESA 2023, 2024 - ICALP 2023 - SIROCCO 2023	
PhD Representative , JHU CS Diversity and Inclusion Committee	Jan-Dec 2022
Worked with department administrators to help start and facilitate the JHU CS PhD Mentor Hour series, where senior PhD students give advice to junior students on a variety of topics in a casual, discussion-based setting	
Co-Organizer , JHU CS Theory Seminar	Jan-May 2022
Networked and liaised with CS theory researchers throughout the country, inviting them to give talks	

Teaching Assistant, Algorithmic Game Theory (JHU) Jan-May 2022

- Held on-demand office hours to assist students with concepts and assignments
- Created rubrics and graded assignments

Peer Mentor, MIT Student Support Services 2017-2019

- Provided guidance, emotional and logistic support for undergraduate students through their leaves of absence (particularly for students leaving for health reasons)

Assistant Educator, Sarah's Science Feb-Aug 2016

- Introduced kindergarten students to basic science concepts through hands-on activities

Organizer and Educator, MIT Educational Studies Program 2014-2016

- Organized the MIT Summer HSSP program, a summer extracurricular education program for over 100 middle and high school students
- Organized the MIT Cascade program, a free after-school high-school program for over 30 low-income Boston-area students
- Interviewed and hired teachers, taught courses, advertised for both programs

TALKS

Light Edge Fault Tolerant Spanners
ICALP 2025, Aarhus, Denmark July 2025

Approximation Algorithms for Optimal Hopsets
ICALP 2025, Aarhus, Denmark July 2025

Relative Survivable Network Design
JHU CS Theory Seminar, Baltimore, MD, USA Nov 2023
Capital Area Theory Seminar (CATS), College Park, MD, USA Nov 2023

Improved Approximations for Relative Survivable Network Design
WAOA 2023, Amsterdam, The Netherlands Sept 2023

Relative Survivable Network Design
APPROX 2022, Online Sept 2022

SKILLS

Programming: Python, Java, $\text{\LaTeX}$
Technical: combinatorics, network/routing optimization, theoretical computer science, discrete mathematics, algorithm development and data structures, graph theory, graph algorithms, mathematical optimization, mixed integer programming, mathematical programming, scheduling optimization, Git, research
Other: Teaching, tutoring, technical writing, project management, collaboration, mentorship, communicating technical content to diverse audiences

AFFILIATIONS

Institute for African American Mentoring in Computing Sciences (IAAMCS), Society for Industrial and Applied Mathematics (SIAM), National Center for Women and Information Technology (NCWIT), Rewriting the Code (RTC)